4MRNA User Manual

Last updated date: 2025-08-18

Overview

① Access the website "Programming Code Generator for 4MRNA".

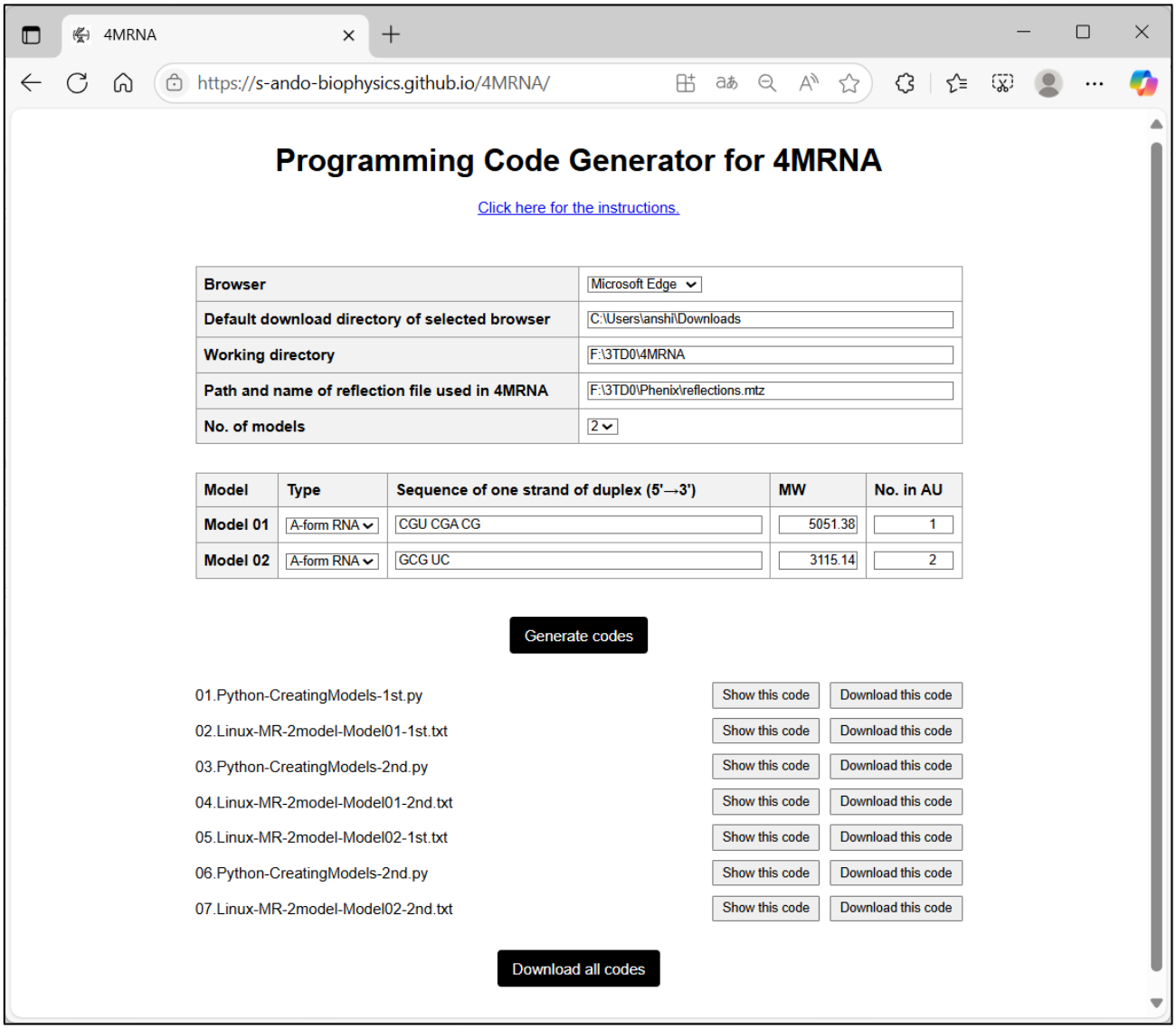
② Enter the required information and complete the table. For details, refer to Detailed Instructions 【Entering Required Information】.

③ Once all required fields are filled in, click the "Generate codes" button. Programming codes corresponding to the entered information will be generated and displayed on the screen.

④ Click the "Download all codes" button to download all the codes.

⑤ The downloaded programming codes are numbered, so execute them in order in your environment. For detailed procedures, refer to Detailed Instructions 【Running the Codes】.

⑥ Once all the codes have finished running, the 4MRNA program is complete. After that, the obtained solution (initial structure) is used for refinement, leading to the determination of the final structure.



Detailed Instructions

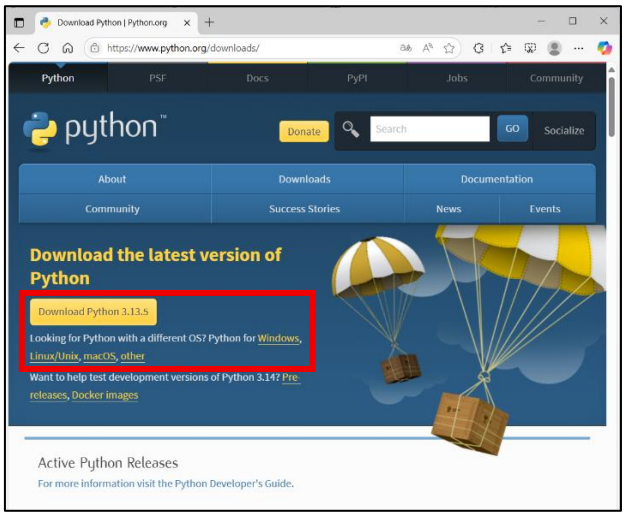
【Preparation and Setup for Using 4MRNA】

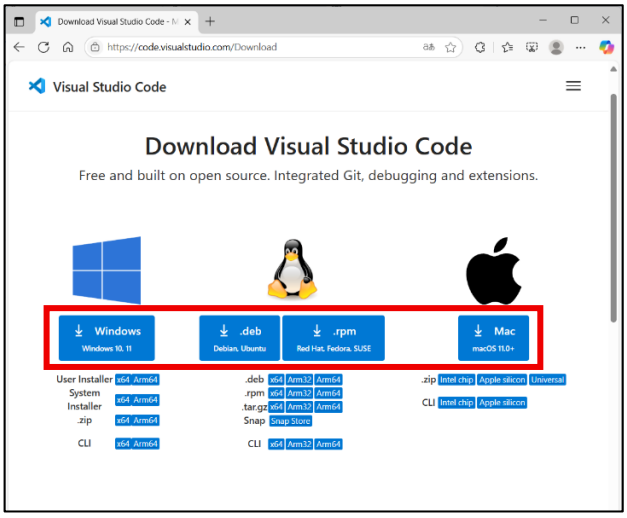
◆ Enable Python

① Install Python. Access the official website, download the installer suitable for your computer's operating system, and run it.

② After installation, open a terminal and execute the following command: pip install selenium pandas.

③ It is recommended (though not mandatory) to download a text editor like Visual Studio Code, which makes it easier to run Python codes.



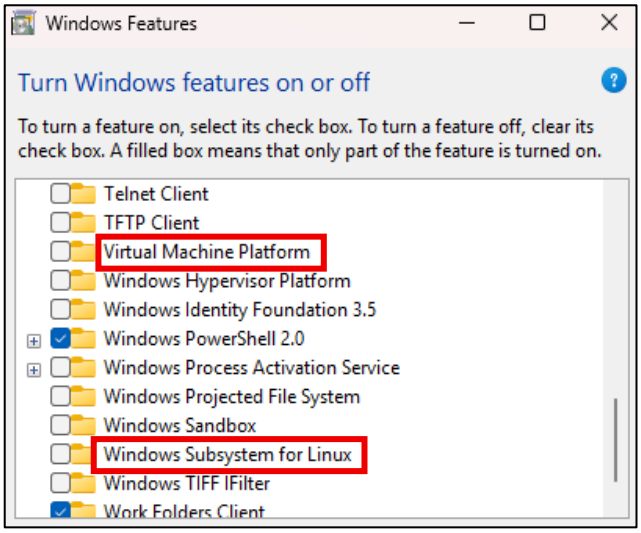


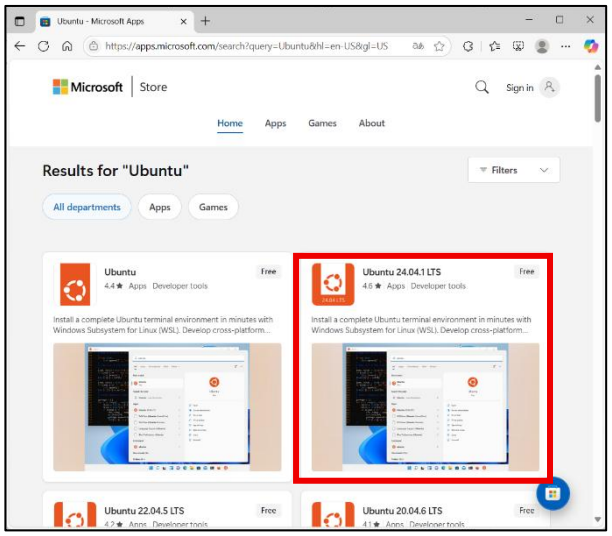
◆ Enable Shell Scripts (Windows Only)

① Enable "Windows Subsystem for Linux (WSL)" and "Virtual Machine Platform". Go to Control Panel > Programs > Programs and Features > Turn Windows features on or off, and check the boxes for "Windows Subsystem for Linux" and "Virtual Machine Platform".

② Download Ubuntu.

③ After the download is complete, launch Ubuntu and run the following commands: sudo apt update and sudo apt upgrade.

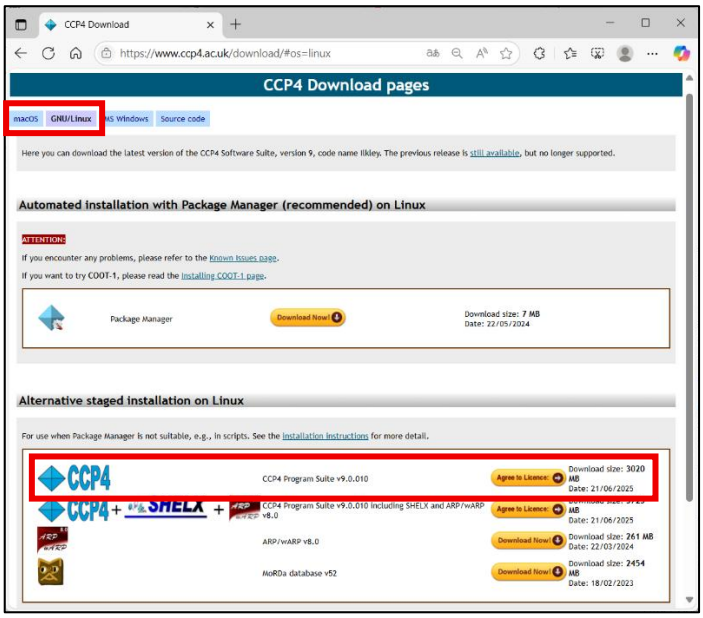




◆ **Enable the Molecular Replacement Software *Phaser***

① Phaser is included in a software suite called CCP4. Visit the official website and download the macOS or Linux version (choose the Linux version even if you're using Windows).

② Once the download is complete, execute the following commands in order to complete the installation. The following commands are for Ubuntu on Windows, but similar commands should be used for macOS and Linux. (These commands assume that the file “ccp4-9.0.010-linux64.tar.gz” has been downloaded to “C:¥Users¥name¥Downloads”. You will need to adjust the version number and folder path as appropriate.)



- sudo su
- cd /usr/local
- mv /mnt/c/Users/name/Downloads/ccp4-9.0.010-linux64.tar.gz .
- gunzip ccp4-9.0.010-linux64.tar.gz
- tar -xvf ccp4-9.0.010-linux64.tar
- cd ccp4-9
- ./BINARY.setup
- exit
- cd /home/name
- vi .bashrc
- source /usr/local/ccp4-9/bin/ccp4.setup-sh
  # Please add to the last line.

◆ **Others**

① If you are using Safari (macOS only), an additional setting is required. Go to Safari Preferences > Develop, and check “Allow Remote Automation”.

② For checking the results of molecular replacement, it is convenient (though not essential) to have spreadsheet software like Microsoft Excel installed.

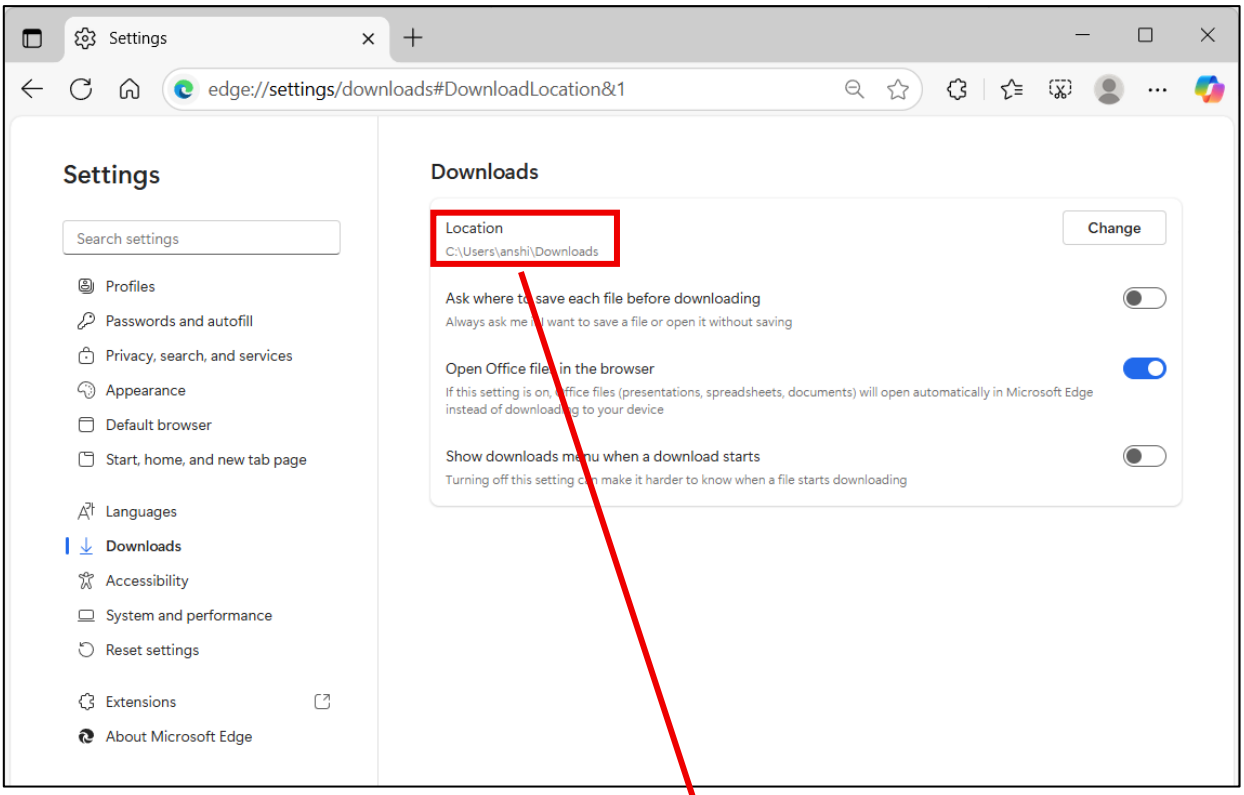
【Entering Required Information (Step②)】

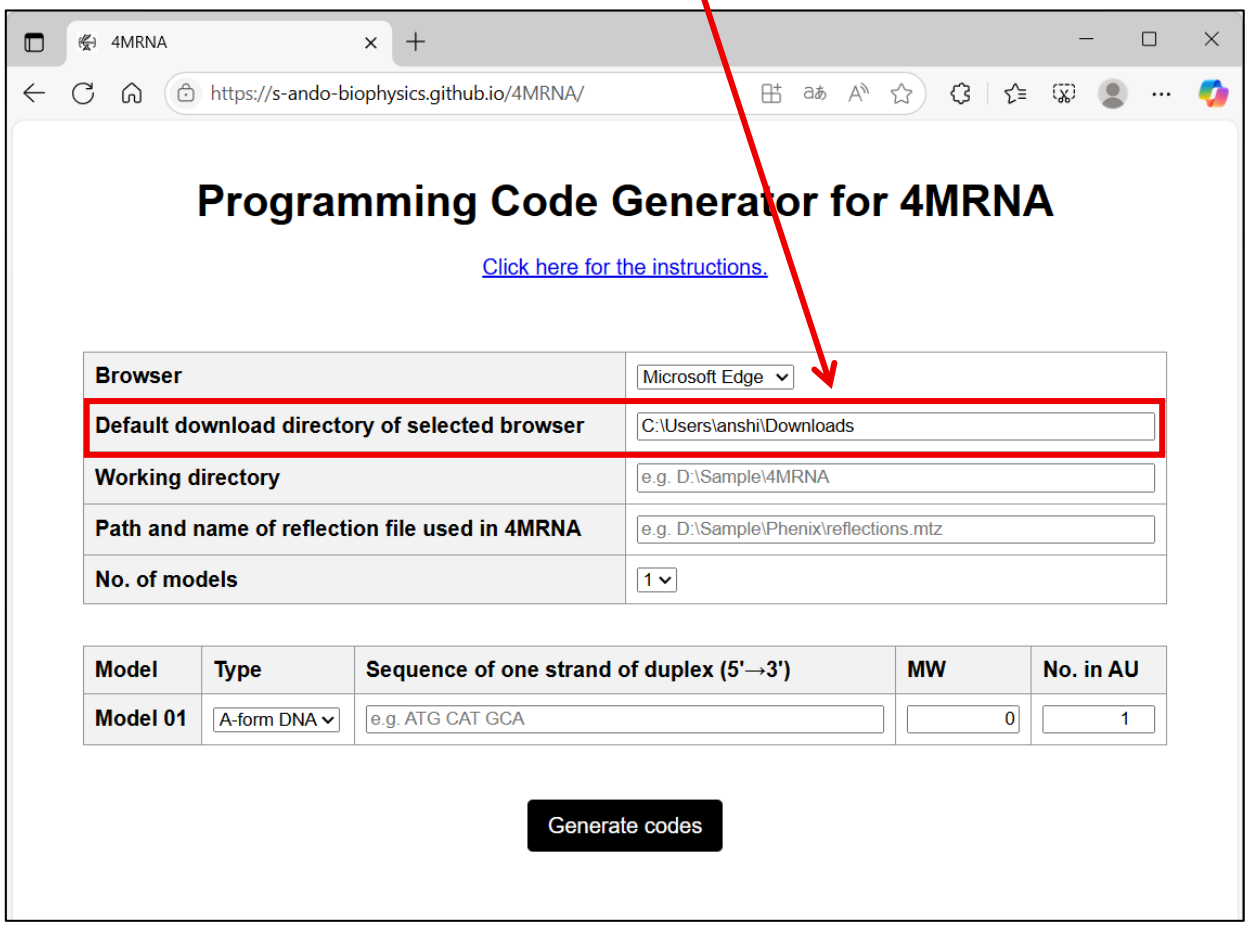
(1)	**Browser**	Select the browser you are using from the following options: Google Chrome, Microsoft Edge, Firefox, Safari.
(2)	**Default download directory of selected browser**	Open the settings of the browser selected in (1), confirm the default download directory, and enter it in this field.
(3)	**Working directory**	Create a directory where 4MRNA will operate (the output files will be saved here), and enter its full path.
(4)	**Path and name of reflection file used in 4MRNA**	Specify the full path and file name of the reflection file used for molecular replacement.

(2) How to Check the "Default download directory of selected browser"

- In most browsers, it can be found under Settings > Downloads.

(e.g.) Windows・Microsoft Edge

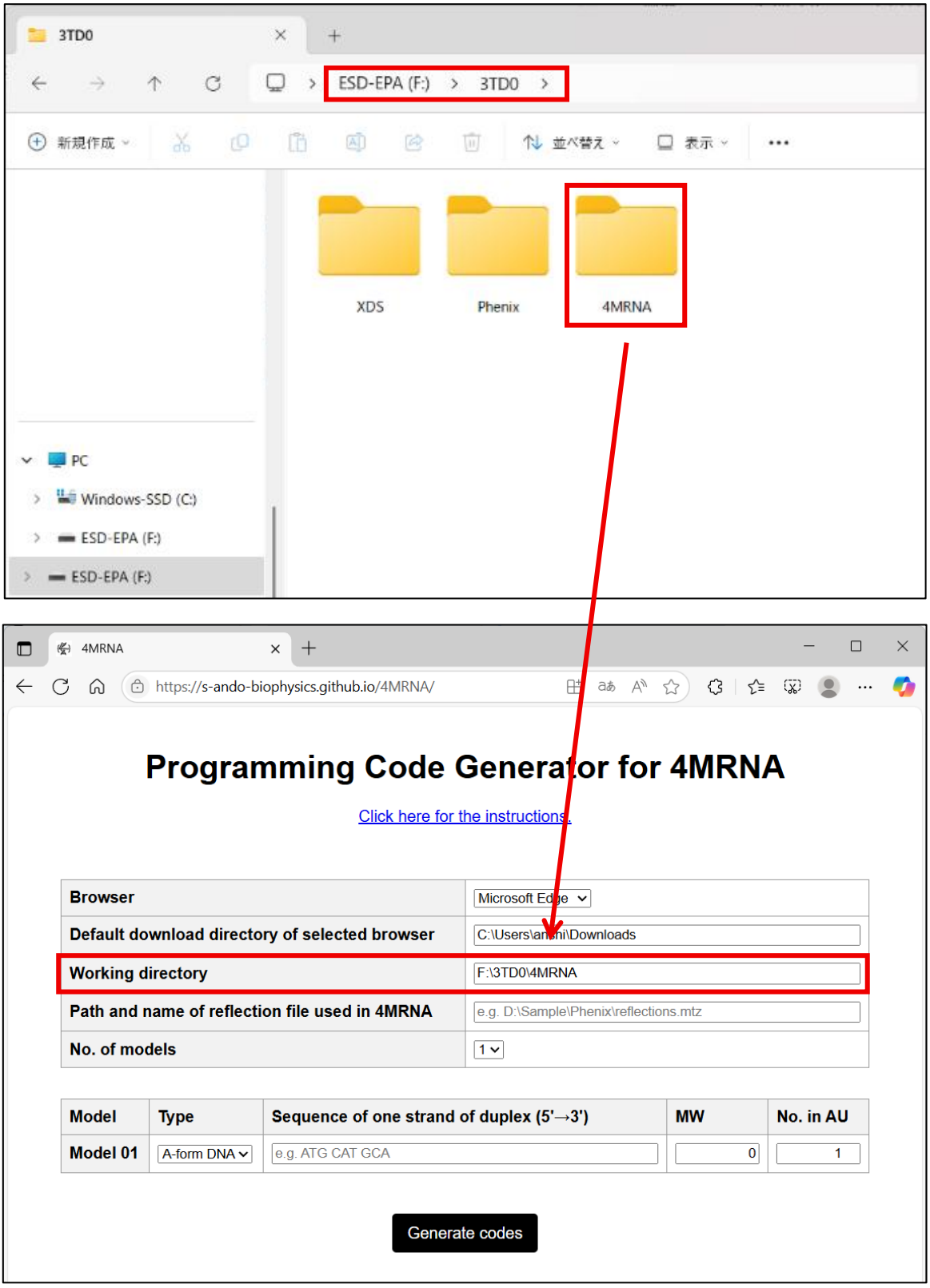




(3) About the "Working directory"

- Create a new folder that 4MRNA will use. Models, parameter files used in model generation, and molecular replacement results will be output to this folder.

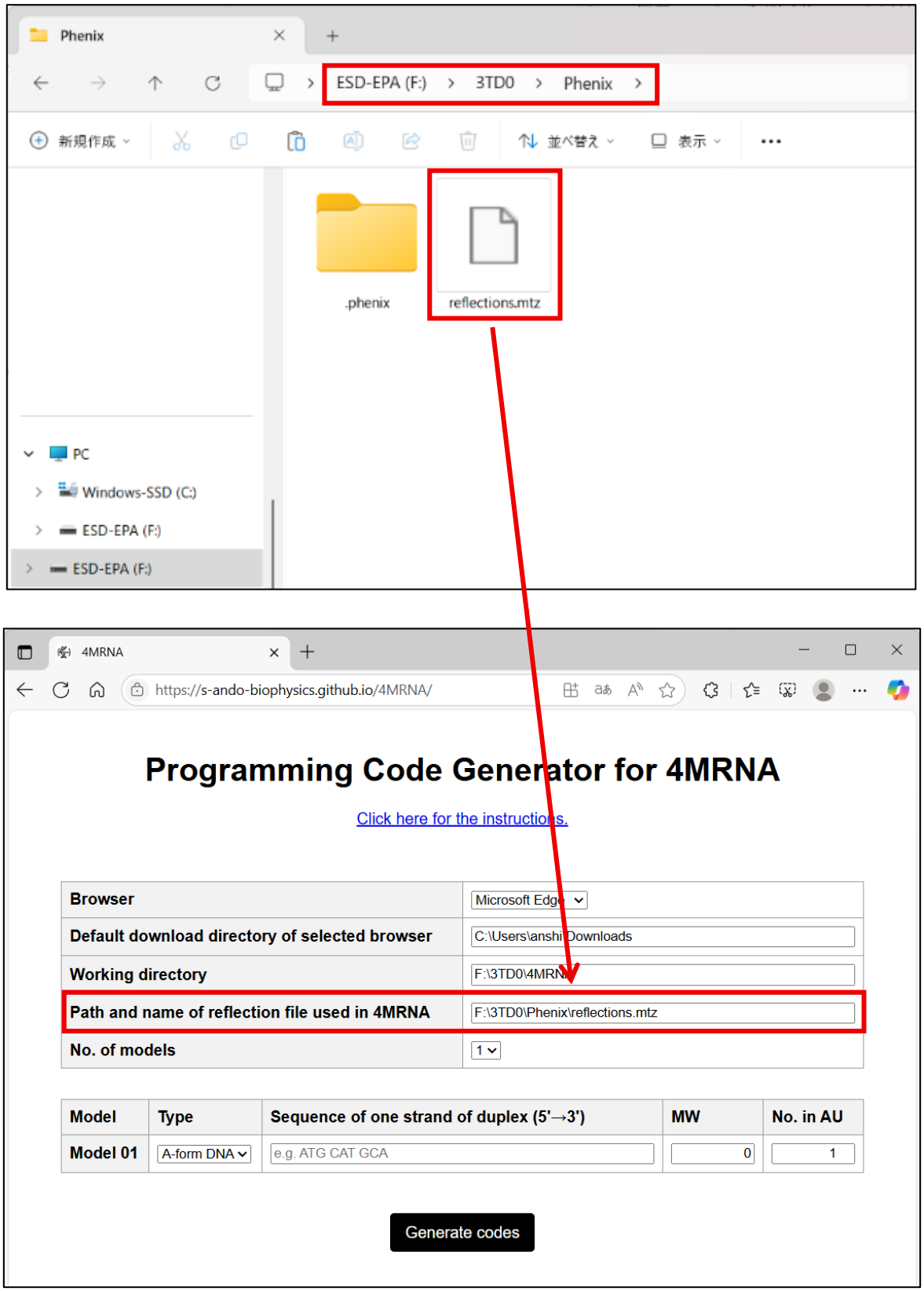
(e.g.) Suppose your sample is named 3TD0, and the X-ray diffraction data is stored in a folder 3TD0 on a USB drive (F:). In that case, create a new folder named “4MRNA” inside “F:¥3TD0”, and input the path "F:¥3TD0¥4MRNA” on the website.



(4) How to Check the "Path and name of reflection file used in 4MRNA"

- You must specify the location and name of the reflection file required for molecular replacement. When using the software *Phenix*, the reflection file is typically named reflections.mtz.

(e.g.) If "reflections.mtz" is in "F:¥3TD0¥Phenix", then input "F:¥3TD0¥Phenix¥reflections.mtz".



(5)	**No. of models**	Select the number of types of search models to be used for molecular replacement.
(6)	**Model Information**	According to the number selected in (5), input fields will appear. For each model, enter the type of nucleic acid duplex, the sequence, and the number of models in the asymmetric unit.

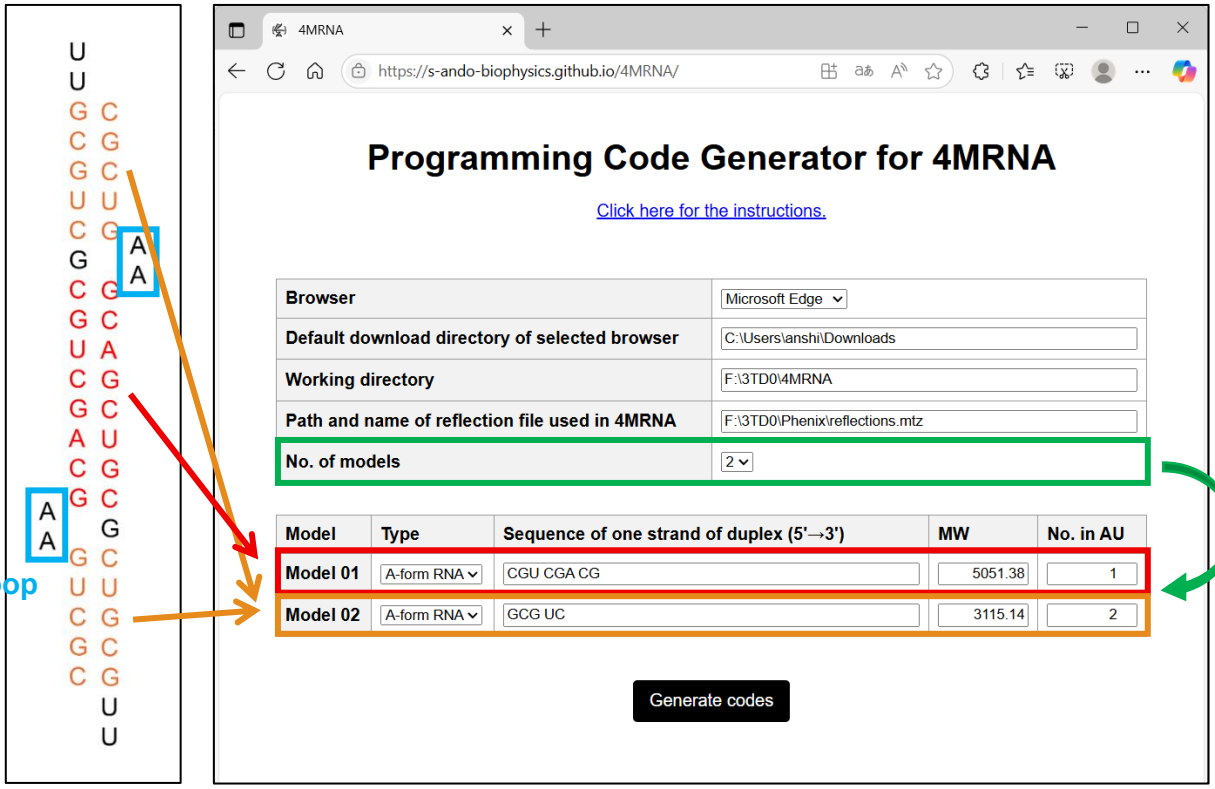
(5) (6) About Search Models Using for Molecular Replacement

① If your sample includes loops in the duplex, divide it into multiple stem regions to avoid the loops and create separate search models for each stem. In that case, the number of models you create (i.e., the number of stem regions) should match the value selected in (5).

② According to the number selected in (5), input fields will appear. For each model (stem), enter the following information.

- **Type**: Select the type of the duplex (stem) from A-form DNA, B-form DNA, or A-form RNA.
- **Sequence of one strand of duplex**: Input the base sequence (5'→3') of one strand of the duplex. The complementary strand will be automatically generated. (Even if mismatches exist, a fully complementary model will be created, which can be corrected later during refinement..)
- **MW**: The molecular weight of the duplex (including complementary strand) will be automatically calculated based on the input sequence.
- **No. in AU**: Enter how many copies of the model exist in the asymmetric unit. This value must be pre-estimated using indicators such as the Matthews coefficient.

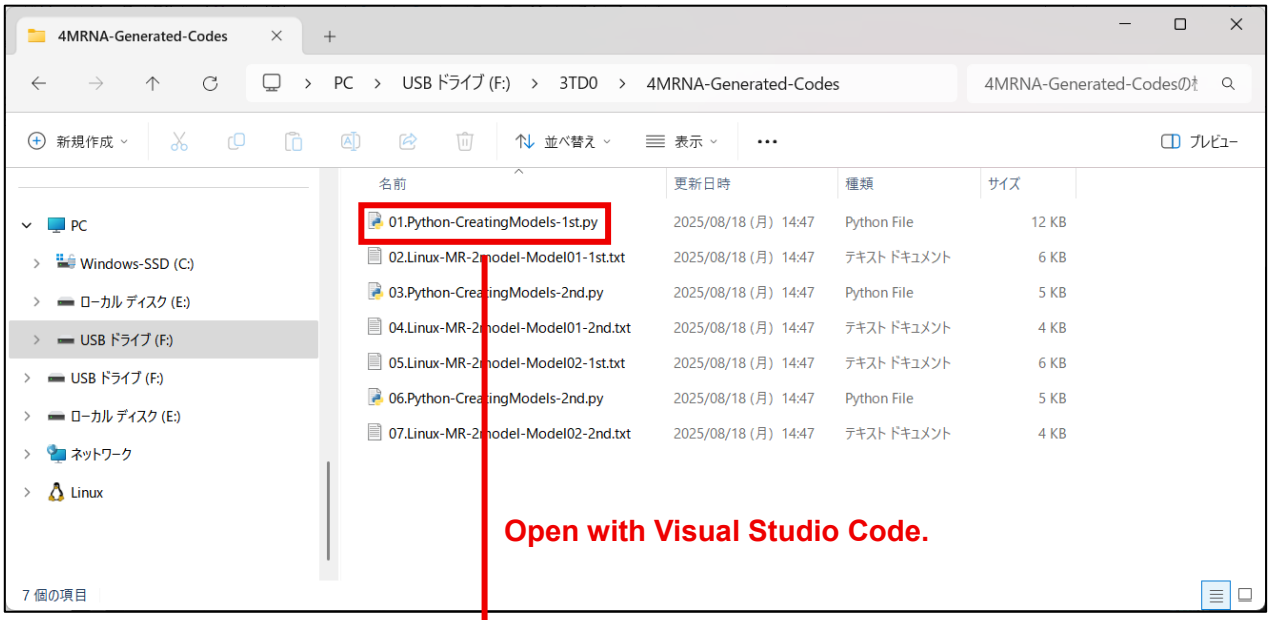
(e.g.) Suppose the structure is expected to contain loops, as shown below. In this case, divide the model into one long stem (red) and two short stems (orange), and input their information accordingly. Assume the Matthews coefficient suggests that one copy of the entire molecule (long stem + 2 short stems + loops) is present in the asymmetric unit.

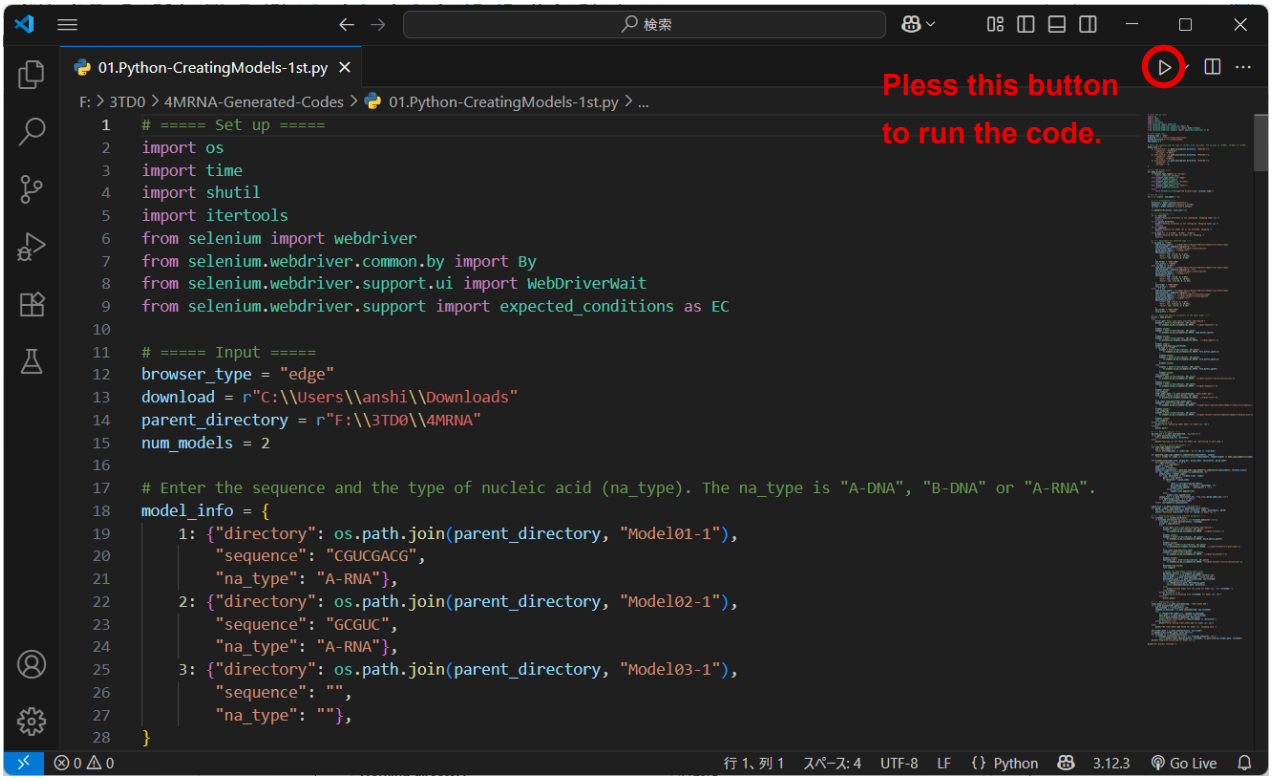


【Running the Codes (Step⑤)】

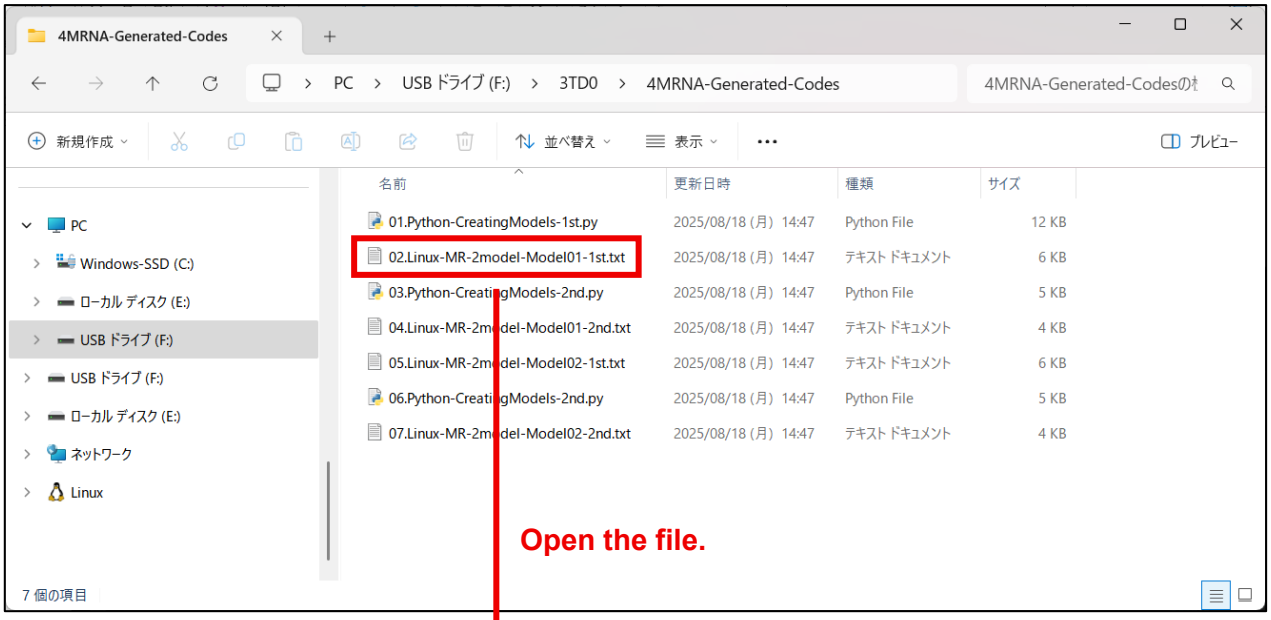
(1)	**Python code**	Among the downloaded files, the ones that start with "Python" and have the .py extension are the Python codes. Open these files using Visual Studio Code or a similar text editor, and run them.
(2)	**Shell script**	Among the downloaded files, the ones that start with "Linux" and have the .txt extension are the shell scripts. Copy the entire content of these .txt files and paste it into Ubuntu (Windows) or a terminal (macOS/Linux).

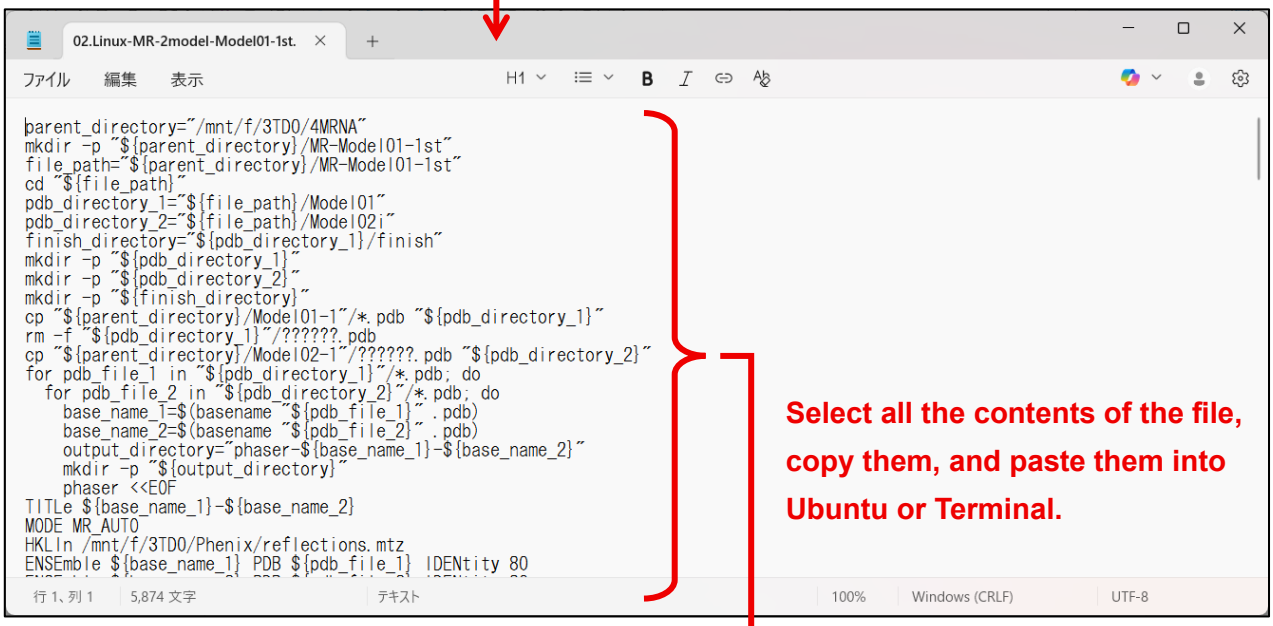
(1) How to Run Python Codes

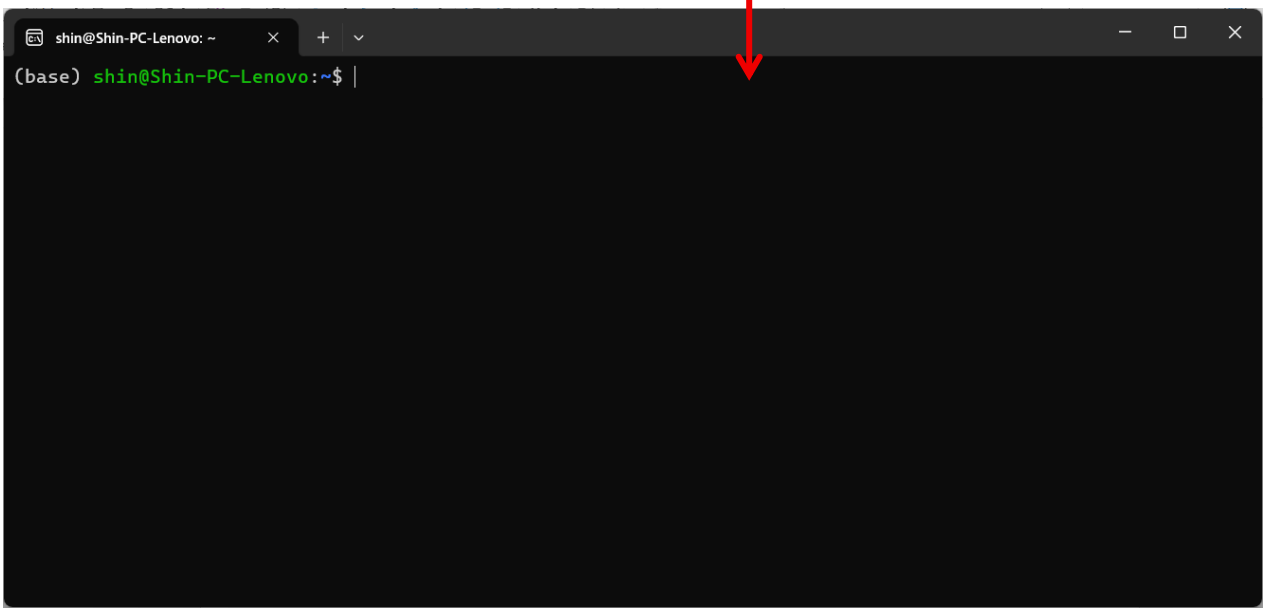




(2) How to Run Shell Scripts







【After Completion of 4MRNA】

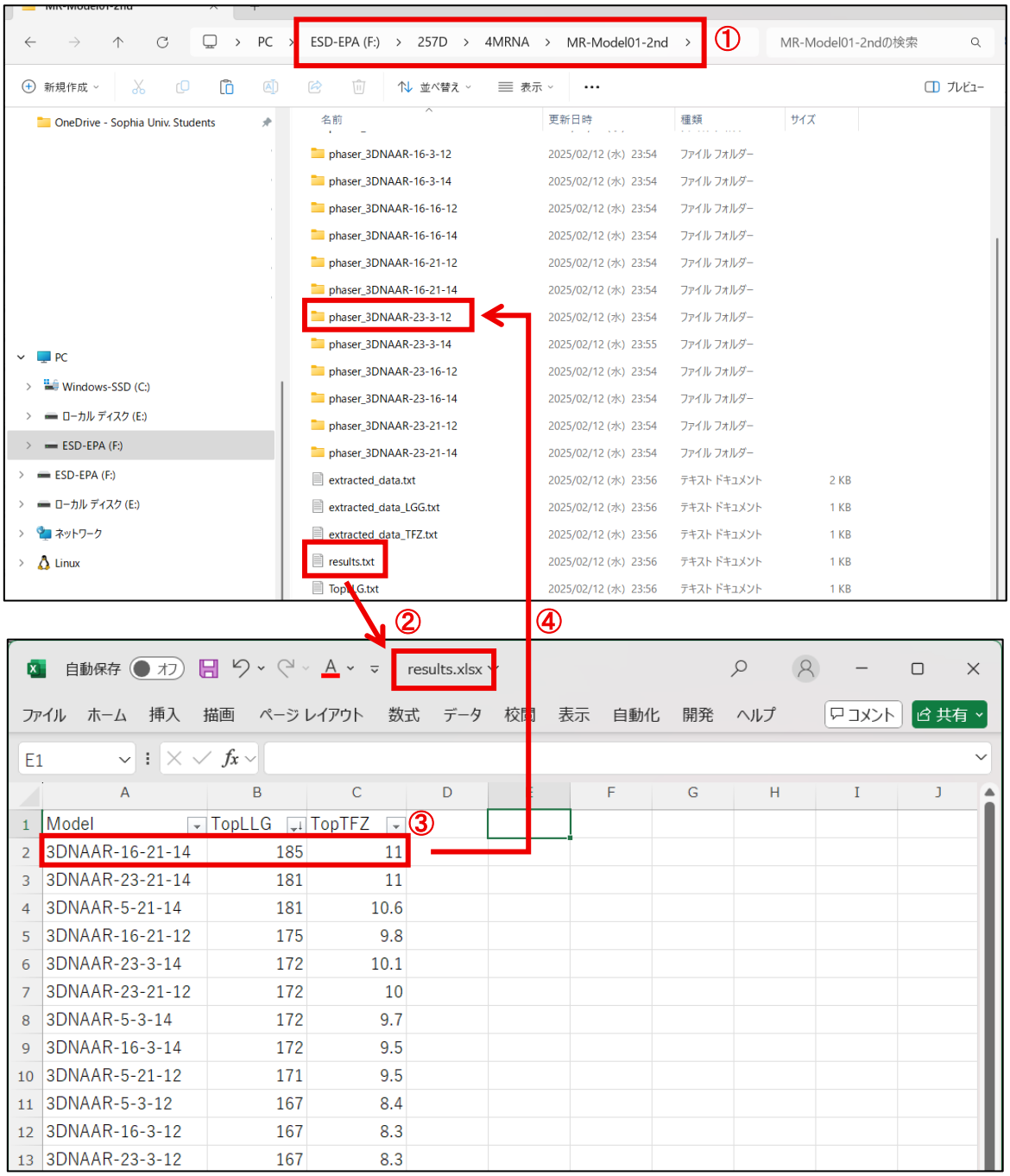
◆ Check the Initial Structure Determined by 4MRNA (Verify the Solution of Molecular Replacement)

① If No. of models = 1 is selected, open the folder “4MRNA > MR-Model01-2nd”. If No. of models = 2 is selected, open the folder “4MRNA > MR-Model02-2nd”. If No. of models = 3 is selected, open the folder “4MRNA > MR-Model03-2nd”.

② Open the file “results.txt” inside the folder. This file contains the molecular replacement statistics (LLG and TFZ). Opening it with a spreadsheet software such as Microsoft Excel makes it easier to compare the statistics (though this is not mandatory).

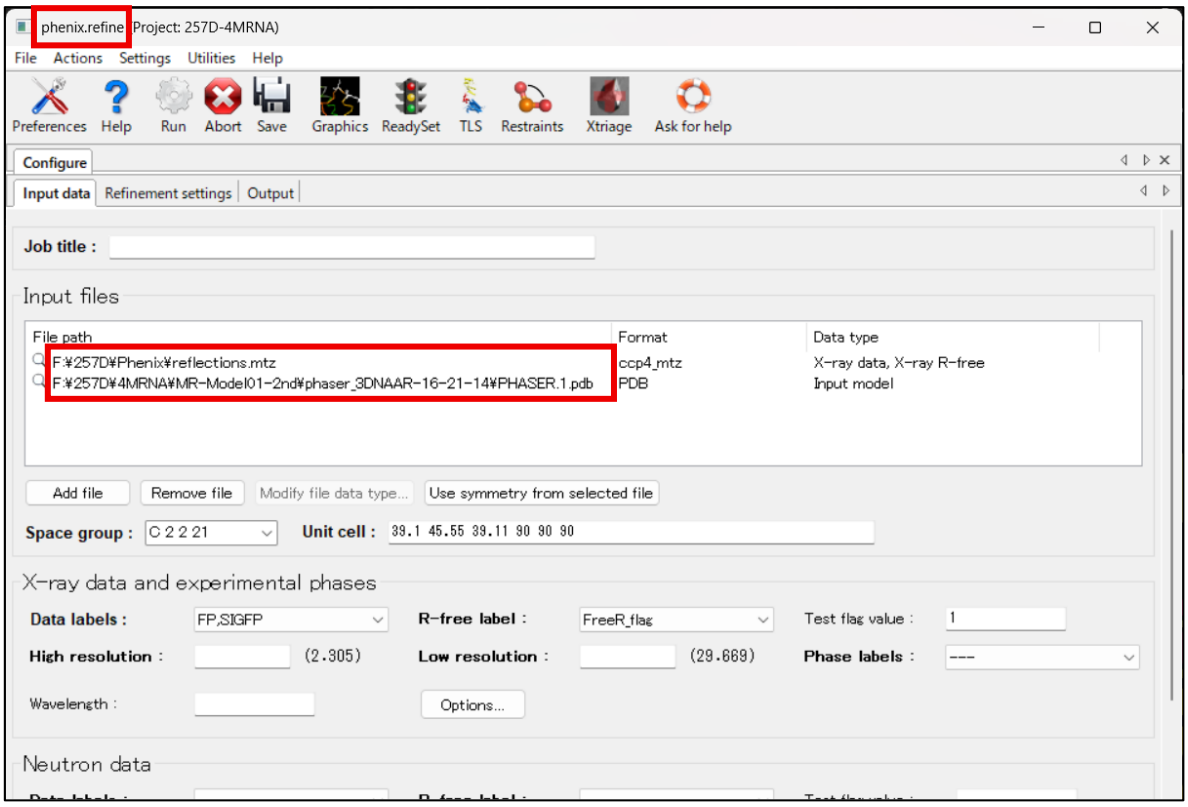
③ Check the statistics and consider the one with the most favorable values as the solution (initial structure). Empirically, if LLG ≥ 100 and TFZ ≥ 7, there is a high probability that the initial structure has been successfully determined

④ Open the folder corresponding to the number selected in step ③. The file “PHASER.1.pdb” inside is the initial structure file.



◆ Refinement Using *phenix.refine* in the Phenix Suite

① Launch *phenix.refine* in Phenix.

② Input the reflection file and the initial structure file (PHASER.1), and adjust the refinement settings as needed.

③ Click the “Run” button to execute the refinement.